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Obstacle Avoidance Path Planning for Robotic Manipulators Based on an Improved Bidirectional RRT* Algorithm

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***Abstract*—To address the problems of low search efficiency and slow convergence speed of the traditional Bidirectional Rapidly-exploring Random Tree Star (Bi-RRT*) algorithm in path planning for six-degree-of-freedom robotic manipulators in complex three-dimensional environments, an improved Bi-RRT* algorithm integrating multiple strategies is proposed. In this algorithm, a dynamic biased sampling strategy based on inter-tree distance awareness was introduced to effectively guide the growth of the dual trees. Concurrently, an optimized method for calculating the inter-tree distance was designed to reduce computational overhead. Finally, path smoothing was performed using pruning and cubic Natural spline curve interpolation to satisfy the kinematic constraints of the robotic manipulator. Simulation and experimental results based on the xArm6 robotic manipulator demonstrate that the improved algorithm significantly reduces planning time, enhances convergence speed, and improves path quality. It also generates a smooth, collision-free trajectory, enabling the robotic manipulator to stably navigate through narrow passages and confirming its practical value for engineering applications.**

Keywords—six-degree-of-freedom robotic manipulators, Obstacle avoidance path planning, Bi-RRT* algorithm, Dynamic biased sampling, Natural spline curve

I. Introduction

Intelligent manufacturing is now an important driver of industrial upgrading. Core factory equipment requires higher levels of automation and intelligence [1, 2]. In cigarette factories, power workshops contain many devices and complex pipelines that need regular inspection for safety. Manual inspection is inefficient, unreliable, and slow in fault detection. Therefore, a robotic inspection system is required.

A system that integrates a mobile platform with a six-degree-of-freedom (6-DoF) manipulator is a representative solution. It combines wide mobility and precise operation. It can move along production lines and perform close-range, high-precision inspections in complex workspaces, enabling real-time monitoring and early warning. At the same time, such tasks impose strict requirements on obstacle avoidance, path optimization, and operational accuracy for the 6-DoF manipulator [3, 4]. Among these, path planning is a key technology that directly affects safety and efficiency and has received extensive attention [5-7].

Mainstream path-planning methods are mainly divided into two categories: graph-search-based methods (e.g., Dijkstra [8], A* [9]) and sampling-based methods (e.g., PRM, RRT). The first group can guarantee optimal or near-optimal paths but has high computational cost in high-dimensional spaces and is mostly used for mobile robots. The second group improves efficiency through random sampling but often yields suboptimal and stochastic solutions. Therefore, improving solution quality and convergence speed remains a focus.

To improve path quality and efficiency, Karaman and Frazzoli [10] proposed the asymptotically optimal RRT* based on random geometric graph theory, which extends the applicability of sampling-based planning. Wang et al. [11] proposed EB-RRT to balance efficiency and optimality by integrating heuristic guidance and boundary-preserving mechanisms. However, RRT and its variants rely on single-tree random expansion, and convergence is slow.

Compared with unidirectional search, bidirectional strategies can greatly accelerate convergence. The Bi-RRT algorithm builds two trees from the start and the goal to cover the configuration space [12]. To further improve convergence, Kuffner et al. [13] proposed RRT-Connect with a greedy connection strategy but without global optimization. Jordan et al. [14] proposed Bi-RRT*, which introduces heuristic techniques to achieve asymptotic optimality with reduced computation. However, convergence is still limited by weak goal-directed sampling.

In this paper, an improved Bi-RRT* is proposed for manipulator obstacle-avoidance path planning. A distance-

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aware dynamically biased sampling mechanism is introduced to enhance goal orientation and accelerate convergence. At the same time, an efficient inter-tree distance computation is designed to reduce computational burden. Finally, path pruning and smoothing are applied to the initial trajectory to generate smooth motion paths suitable for industrial inspection tasks.

II. Kinematic Modeling of Robotic Manipulator Using D-H Method

Robotic manipulators are now an important part of modern industrial systems. In this study, the UFACTORY xArm6, a typical six-degree-of-freedom (6-DoF) serial arm, is used as the research platform. According to classical kinematics, the Denavit–Hartenberg (D-H) method is adopted to describe the geometric relationships among links and to build the kinematic model [15]. At the same time, Figure 1 shows the coordinate frames assigned to each xArm6 link under the standard D-H convention, which clearly present the relationships among joint axes and link parameters [16].

The standard D-H parameters of the xArm6 are given in Table 1. Here, θi denotes the i-th joint angle and is the variable for revolute joints. The initial configuration sets all joint angles to zero. The term $a_{i\text{-}1}$ is the link length, defined as the distance between the $Z_{i\text{-}1}$ and Z_i axes along their common normal. The term $\alpha_{i\text{-}1}$ is the link twist, i.e., the rotation about the $X_{i\text{-}1}$ axis from $Z_{i\text{-}1}$ to Z_i. The term d_i is the link *offset*, and *offset* denotes a constant joint-angle shift.

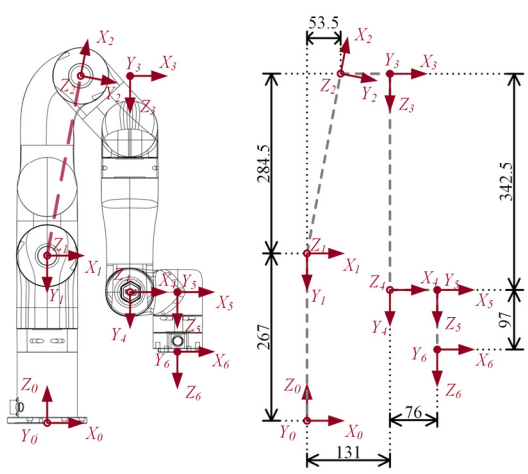

Figure 1. Definition of the joint coordinate systems for xArm6 robotic manipulator.

TABLE 1. STANDARD D-H PARAMETERS FOR XARM6 ROBOTIC MANIPULATOR.

Joint i	a_{i-1} / mm	α_{i-1} / rad	d_i / mm	θ_i / rad	offset/ rad
1	0	$-\pi/2$	267	θ_1	0
2	289.5	0	0	θ_2	-1.3849
3	77.5	$-\pi/2$	0	θ_3	1.3849
4	0	$\pi/2$	342.5	θ_4	0
5	76	$-\pi/2$	0	θ_5	0
6	0	0	97	θ_6	0

III. Improved Bi-RRT* Algorithm

A. Basic Principle of the Improved Bi-RRT* Algorithm

Sampling-based motion planning is now an important part of robotic navigation in large-scale 3D spaces. Bidirectional RRT* (Bi-RRT*) is a typical representative of this class. In practice, search efficiency is a key issue when the space is large. In this paper, the standard Bi-RRT* is enhanced by a distance-aware dynamic bias and a multi-level optimization mechanism. At the same time, the algorithm monitors the minimum distance between the two trees and modifies sampling according to this distance to guide their mutual growth. The main components include an intelligent distance update, an optimized connection check, and a fast biased sampling scheme. As the trees expand, the distance-aware mechanism coordinates their bidirectional growth and accelerates path connection.

B. Distance-Aware Dynamic Biased Sampling Strategy

In the path planning problem, the three-dimensional search space can be described as a bounded convex set

$$X = [0, L_x] \times [0, L_y] \times [0, L_z] \subset R_3 \tag{1}$$

Here, $L = (L_x, L_y, L_z)^T$ represents the spatial boundary vector. During the k-th iteration, the start tree and goal tree are respectively defined as

$$T_s^k = \{(x_i^s, p_i^s, c_i^s)\}_{i=1}^{n_s^k}, T_g^k = \{(x_i^g, p_i^g, c_i^g)\}_{i=1}^{n_g^k}$$

where x_i^s, x_i^g denote the three-dimensional coordinates of the nodes, p_i^s, p_i^g and p_i^g represent the indices of the parent nodes, and c_i^s and c_i^g are the cumulative path costs from the root node to the current node.

The minimum Euclidean distance between the two trees can be expressed as:

$$d_{\min}^k = \min_{1 \le i \le n_s^k, 1 \le j \le n_g^k} \left\| x_i^s - x_j^g \right\|_2 \tag{2}$$

and the corresponding pair of nearest nodes is given by:

$$(x_{s*}^k, x_{g*}^k) = \arg\min_{i,j} \left\| x_s^i - x_g^i \right\|_2 \tag{3}$$

To eliminate the influence of spatial scaling on parameter computation, a normalized distance is introduced as

$$\hat{d}_k = \frac{d_{\min}^k}{\|L\|_2} \tag{4}$$

The algorithm uses an adaptive bias strategy and dynamically balances exploration and exploitation. At the same time, Figure 2 illustrates the dynamic biased sampling and the node expansion process. When the two search trees are far apart, the goal bias is reduced, and global exploration is prioritized to find better connection regions. As the trees converge, the goal bias is increased, and the focus shifts to exploiting the current

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promising paths to accelerate the inter-tree connection. This mechanism is implemented by an exponential decay function that varies with the distance between the two trees, given by:

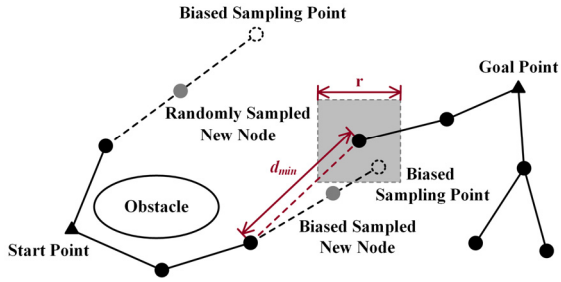

Figure 2. Schematic of dynamic biased sampling.

$$P_{bias}^{k} = P_0 + (P_{\max} - P_0) \cdot e^{-\alpha \partial_k} \tag{5}$$

Here, P_0=0.1 represents the base bias probability, which ensures the exploration capability of the algorithm, while $P_{\max}$=0.25 denotes the maximum bias probability, preventing excessive bias that may lead to premature convergence or local optima. The parameter α=2 is the decay coefficient, controlling the sensitivity of the bias strength to the variation of the inter-tree distance.

The expansion strategies of the start tree and goal tree are approximately symmetric. For the expansion of the start tree, the new sample point x_{sample}^{k} follows a probabilistic distribution as:

$$x_{sample}^{k} \sim \begin{cases} \psi(x_k^{g^*}, r), & P < P_{bias}^{k} \\ \xi(X), & P \geq P_{bias}^{k} \end{cases} \tag{6}$$

where $\xi(X)$ denotes the uniform distribution over the search space, and $\psi(x_k^{g^*}, r)$ represents the local biased distribution centered at the nearest goal-tree node.

The local biased region adopts a truncated cubic distribution instead of a spherical one for simplicity, which can be expressed as:

$$\psi(x_k^{g^*}, r) = \xi([x_k^{g^*} - r, x_k^{g^*} + r] \cap X) \tag{7}$$

Where $r = 0.15 \cdot \min(L_x, L_y, L_z)$ 。

C. Optimized Inter-Tree Distance Computation

The computational burden of dynamic biased sampling is a key issue in large-scale 3D spaces. In this paper, an efficient method is used to estimate the minimum distance between the two trees. Instead of recomputing Euclidean distances for all nodes at each iteration, a random subset sampling and an adaptive update are adopted. This design balances distance accuracy and computational cost. When the number of nodes is very large, only a small subset of node pairs is checked, and the process terminates early when the minimum distance satisfies a preset threshold. At the same time, an adaptive update adjusts the distance update interval according to the rate of change, and unnecessary updates are avoided when the distance is stable. Figure 3 shows the optimized workflow, where the update frequency and computation cost are significantly reduced while accuracy is maintained.

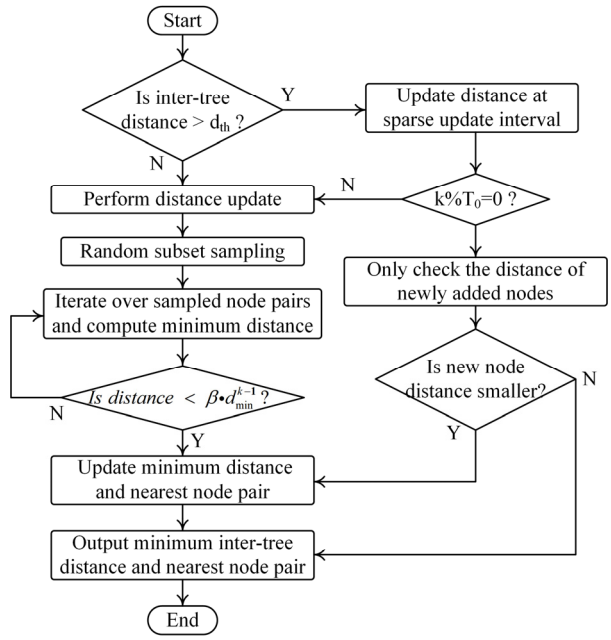

Figure 3. Optimized inter-tree distance computation workflow.

D. Schematic of path curve optimization based on cubic spline interpolation.

Manipulator trajectory smoothing is now an important part of motion planning. In this study, the trajectory is optimized by cubic natural spline interpolation. After redundant nodes are removed, the spline enforces second-order continuity. In this way, sharp corners and discontinuities are avoided, and the trajectory is smooth and stable for safe execution. At the same time, Figure 4 shows the spline-based path optimization: discrete path points are taken as input, and a smooth curve is generated to connect the remaining path points, eliminate sharp turns, and improve continuity. As a result, the smoothness and feasibility of the planned trajectory are significantly improved.

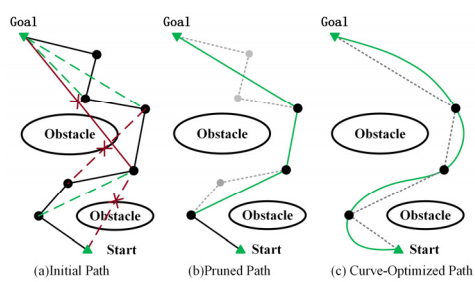

Figure 4. Schematic of path curve optimization.

IV. Experiments and Data Analysis

A. Three-Dimensional Path Planning Simulation

Three-dimensional path planning is now an important part of robotic navigation. To evaluate the improved algorithm, two MATLAB test environments with different complexity are constructed: a simple sparse-obstacle scene and a complex maze-like dense-obstacle scene. Because the methods are sampling-based and therefore random, the original and the improved algorithms are run 100 times in each environment to ensure statistical validity.

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(1) Path Planning in Sparse-Obstacle Environment

The planning space is 150 x 150 x 100; the start point is (10, 10, 10) and the goal is (120, 120, 80). The search radius is 10 and the step size is 5. Key parameters of the improved algorithm are a base bias probability of 0.01, a maximum of 0.25, a decay coefficient of 2, a sampling size of 50, and an improvement threshold factor of 0.8. This configuration serves as a baseline to verify performance gains and to ensure that the optimization strategy does not introduce extra computational cost. At the same time, Figure 5 shows the expansion process of both algorithms in this simple scene.

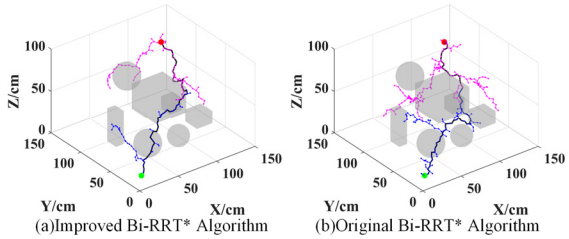

Figure 5. 3D planning result in simple scenario.

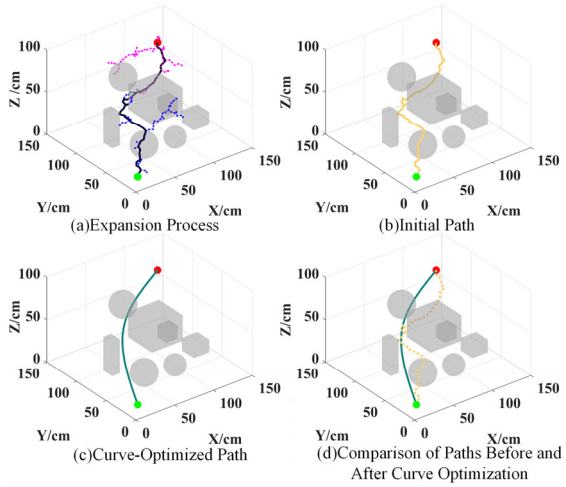

Figure 6. Improved algorithm path planning process in simple scenario.

Figure 6 presents the planning process and the optimized path of the improved algorithm in a simple environment. Part (a) shows the tree expansion; part (b) shows the initial plan with redundant nodes; part (c) presents the pruned and smoothed trajectory, which is more continuous and natural; and part (d) compares the initial and optimized paths generated by the optimization module, showing improvements in both path length and quality.

In this environment, the improved planner achieves a 4.65% reduction in path length (mean ± SD 247.77 ± 16.23 cm versus 259.85 ± 21.39 cm), a 38.79% reduction in planning time (0.46 ± 0.17 s versus 0.76 ± 0.32 s), and a 34.77% reduction in iterations (125.35 ± 33.35 versus 192.17 ± 54.56) compared with the original method. These results indicate a clear improvement in planning efficiency. The variability is also reduced, as indicated by decreases in standard deviation of 24.12% for path length, 46.88% for planning time, and 38.89% for iterations. At the same time, no additional computational burden or time is introduced. Therefore, the proposed optimization strategy provides efficient and lightweight planning performance.

(2) Maze-Like Dense-Obstacle Path Planning

The planning space is 150 x 150 x 100; the start point is (10, 30, 10) and the goal is (125, 125, 80). The search radius is 10 and the step size is 5. Key parameters of the improved algorithm are a base bias probability of 0.01, a maximum of 0.25, a decay coefficient of 2, a sampling size of 50, and an improvement threshold factor of 0.8. This complex scene requires strong search capability and is used to evaluate exploration ability and convergence speed under challenging constraints. At the same time, Figure 7 shows the expansion processes of both algorithms in this complex scene.

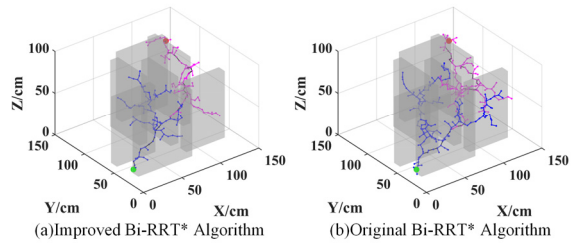

Figure 7. 3D planning result in complex scenario.

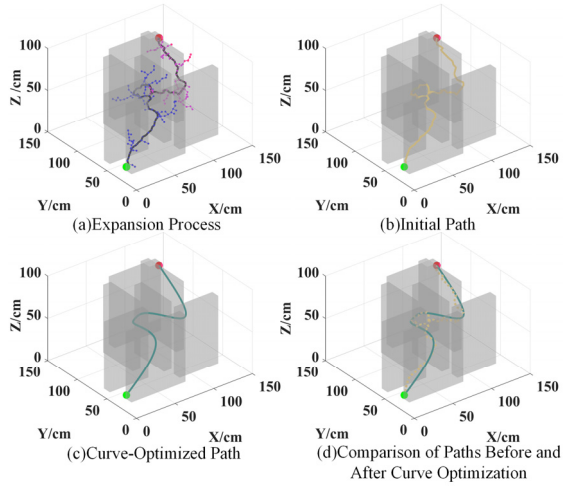

Figure 8. Improved algorithm path planning process in complex scenario.

Figure 8 further presents the planning procedure of the improved algorithm in a complex environment. Similar to Figure 6, it shows the complete process from exploration to optimization in four subfigures.

In this environment, the improved planner achieves a 3.11% reduction in path length (mean ± SD 261.60 ± 21.35 cm versus 270.01 ± 26.73 cm), a 21.13% reduction in planning time (0.75 ± 0.56 s versus 0.95 ± 0.69 s), and a 22.68% reduction in iterations (158.10 ± 64.63 versus 204.47 ± 75.66) compared with the original method. These results indicate that the proposed multi-strategy fusion effectively guides the search direction and limits invalid exploration, leading to improved planning efficiency while maintaining path quality. The variability is also reduced, as indicated by decreases in standard deviation of 20.13% for path length, 18.84% for planning time, and 14.57%

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for iterations. Therefore, the proposed multi-strategy fusion provides efficient planning performance in this environment.

B. 6-DOF Robotic manipulator Obstacle Avoidance Path Planning

Manipulator obstacle-avoidance planning is now an important part of real robotic operations. In this study, the MATLAB Robotics Toolbox is used to import the D-H parameters of the xArm6, and the improved algorithm is validated for stability and effectiveness in a realistic setting. The planning space is 150 x 150 x 100 and contains four rectangular obstacles arranged as a 3D equipment layout in a narrow passage. The start point is (40, 0, 40) as the standby position, and the goal is (100, 80, 55) as a critical monitoring point on the opposite side. This configuration emulates an inspection task in the tight confines of a cigarette factory power workshop and is used to evaluate displacement planning and obstacle avoidance under strong spatial constraints.

At the same time, Figure 9 shows the manipulator poses along the trajectory generated by the improved algorithm as it passes through the narrow area and reaches the monitoring zone. The experiments indicate that the manipulator moves smoothly in the workspace and avoids obstacles successfully. These results demonstrate the feasibility and practical applicability of the improved algorithm in real robotic scenarios.

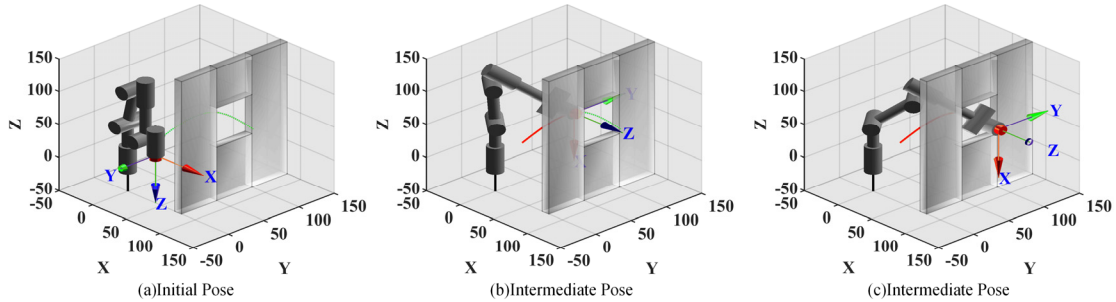

Figure 9. Robotic manipulator poses during trajectory execution.

V. Conclusion

Six-degree-of-freedom manipulator path planning in complex 3D spaces is now an important part of robotic applications. The standard Bi-RRT* suffers from high sampling blindness and slow convergence. In this study, an improved Bi-RRT* with multi-strategy optimization is proposed. The method integrates distance-aware dynamic biased sampling, random subset sampling with early termination, and adaptive parameter adjustment. In this way, sampling efficiency and convergence speed are improved and the computational burden is reduced. Experiments show that in a sparse-obstacle environment, path length, planning time, and iterations are reduced by 4.65%, 38.79%, and 34.77%, and in a maze-like dense-obstacle environment by 3.11%, 21.13%, and 22.68%. The standard deviations also decrease by 24.12%, 46.88%, and 38.89% in the sparse case and by 20.13%, 18.84%, and 14.57% in the dense case, indicating more stable and predictable convergence without extra computational burden or time. The algorithm generates shorter, smoother paths and improves both planning efficiency and path quality.

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